Jiangsu Jingchuang Electronics Co., Ltd.

Temperature Controller EK3030E

MODBUS PROTOCOL

Compiler: Checker: Approver:

EK3030E-MODBUS PROTOCOL

1. INTRODUCTION

1.1 Modbus Protocol

The controller adopts the communication protocol of MODBUS-RTU slave mode. It supports individual communication mode rather that broadcast mode. (The master can address individual slaves or can initiate a broadcast message to all slaves. Slaves return a message called a “response” to queries that are addressed to them individually. Responses are not returned to broadcast queries from the master.)

1.2 Serial Configuration

Serial interface	TTL
Baud rate	9600
Data length	8 bit
Parity	None
Stop bit	1 bit

Note: TTL serial interface is adopted. Communication adapter card is optional and can be converted into RS485 communication interface.

1.3 Message Frame

Start	Device address	Function code	Data field	CRC16		End
≥4ms interval	1 byte	1 byte	N byte	Low-order byte	High-order byte	≥4ms interval

Start/End: The controller monitors the data from communication port continuously. If the silent interval (no data) exceeds that of 3.5 character times, the received data is considered as a new data frame. In this protocol, the silent interval is defined to be 4ms (With Baud rate 9600, 4ms>3.5 character times).

Device address: the first byte transmitted, indicating the slave device will receive the message from the master. A master addresses a slave by placing the slave address in the address field of the message. Every slave has only one address. The slave that only has correct address can respond. When the slave sends its response, it places its own address in this address field of the response to let the master know which slave is responding.

Function Code: the second byte transmitted. When a message is sent from a master to a slave device the function code field tells the slave what kind of action to perform. Modbus defines the function codes in the range of 1-127. If the slave sends a function code with top digit 1 (function code>127), it does not respond or an error occurs.
The controller only uses part of function codes. 0x03 reads holding register (to read current binary value in one or multiple holding registers). 0x06 writes single holding register (to write a specific binary value into a holding register). 0x2B reads device ID (to read the device ID related to device's physical and functional description).

Error checking field: The master and slave devices can use check code to discriminate whether received information is wrong. Sometimes, information will change slightly in transmission due to electronic noise or other interference. Error checking code can ensure the master and slave devices do not handle the error information in transmission, improving system safety and efficiency. Error checking method CRC16 is adopted.
The C Programming Language function generated by CRC16 is as follows: unsignedchar *puchMsg; // points to the pointer used to generate CRC binary data message buffer. unsigned shortusDataLen; // the number of bytes in message buffer.

```
unsignedshort CRC16 ( puchMsg，          usDataLen )
{
        unsignedchar uchCRCHi= 0xFF ;            //CRC high-order byte initializing
        unsignedchar uchCRCLo = 0xFF ;           //CRC low-order byte initializing

        unsigneduIndex ;                         //CRC query table index
        while (usDataLen--)                      // finish the whole message buffer
        {
                uIndex = uchCRCLo^ *puchMsgg++ ; // calculate CRC
                  uchCRCLo = uchCRCHi ^ auchCRCHi[uIndex]; uchCRCHi
                  = auchCRCLo[uIndex] ;
        }crcl
        return (uchCRCHi << 8 |uchCRCLo);
}
```

1.4 Error Message

Device address	Function code	Exception code	CRC16	
1 byte	1 byte	1 byte	Low-order byte	High-order byte

Address code:
The address code sent by the slave indicates the slave address.

Function code:

In exception response, function code sent by the server MSB = 1. E.g. 0x03 returns an error message with function code 0x83.

Exception code:

Exception code	Name	Meaning
0x01	Invalid function	For the server (or slave station), a function code unavailable was requested, different from 0x03, 0x06, 0x2B in this controller.
0x02	Invalid address	For the server (or slave station), an address unavailable was requested.
0x03	Invalid data value	For the server (or slave station), a value unavailable was requested.

2. FUNCTION CODE

2.1 0x03 Read Holding Register

The master sends a command:

Device address	1 byte	Slave address
Function code	1 byte	0x03
Start address	2 bytes	0x0100 ~ 0x0102， 0x0400 ~ 0x042C， 0x0800 ~ 0x0802， 0x0A00
Number of registers	2 bytes	1-10

The slave responds:

Device address	1 byte	Slave address
Function code	1 byte	0x03
Number of bytes	1 byte	2*N (N represents the number of registers)
Register value	N*2 bytes	

Error

Device address	1 byte	Slave address
Function code	1 byte	0x83
Exception code	1 byte	01 or 02 or 03

E.g. the master address is 1 and requests the slave to read the command of register 108-110, so it sends 01 03 00 6B 00 03 XX YY, wherein, 01: slave address; 03: function code; 00 6B: start address; 00 03: the number of registers; XX YY: CRC check code.

The slave responds: 01 03 06 02 2B 00 00 00 64 XX YY; wherein, 01: slave address; 03: function code; 06: the number of data bytes sent; 02 2B 00 00 00 64: the value in register 108-110; XX YY: check code.

If error data is received, the slave will return data 01 83 01 XX YY; wherein, 01: slave address; 83: exception code; 01: exception code; XX YY: checksum.

2.2 0x06 Write Single Holding Register

The master sends a command:

Device address	1 byte	Slave address
Function code	1 byte	0x06
Register address	2 bytes	0x0400 ~ 0x042C， 0x0A00
Register value	2 bytes	0x0000-0xFFFF

The slave responds:

Device address	1 byte	Slave address
Function code	1 byte	0x06
Register address	2 bytes	0x0400 ~ 0x042C， 0x0A00
Register value	2 bytes	0x0000-0xFFFF

Error

Device address	1 byte	Slave address
Function code	1 byte	0x86
Exception code	1 byte	01 or 02 or 03

E.g. the master address is 0x01 and requests the slave to write the data 00 03 in the command of register 2, so it sends 01 06 00 02 00 03 XX YY, wherein, 01: slave address; 06: function code; 00 02: register address; 00 03: the data to write in; XX YY: CRC check code.

The slave responds: 01 06 00 02 00 03 XX YY; wherein, 01: slave address; 06: function code; 00 02: register address; 00 03: the data to write in; XX YY: CRC check code.

If data error or transmission exception occurs, 01 86 02 XX YY is sent.

2.3 0x2B Read Device ID

The master sends a command:

Address code	1 byte	Slave address
Function code	1 byte	0x2B
MEI type	1 byte	0x0E
Device ID	1 byte	0x01 (basic device ID)
Object ID	1 byte	0x00

Basic device ID: company name, product model and revision No.

Object ID:

Object ID	Object name/Description	Type	Object ID value
0x00	Company name	ASCII character string	'elitech'
0x01	Product model	ASCII character string	'EK3030E'
0x02	Main revision No.	ASCII character string	'V1.0'

The slave responds:

Address code	1 byte	Slave address
Function code	1 byte	0x2B
MEI type	1 byte	0x0E
Device ID	1 byte	0x01 (Basic device ID)
Consistency level	1 byte	0x01 (Basic device ID, only stream access)
More	1 byte	0x00
Id Next object ID	1 byte	0x00
Number of objects	1 byte	0x03
Object ID	1 byte	0x00
Object length	1 byte	0x07
Object value	7 bytes	'elitech'
Object ID	1 byte	0x01
Object length	1 byte	0x07
Object value	7 bytes	'EK3030E'
Object ID	1 byte	0x02
Object length	1 byte	0x04
Object value	4 bytes	'V1.0'

Error

Device address	1 byte	Slave address
Function code	1 byte	0xAB (0x2B + 0x80)
Exception code	1 byte	01 or 02 or 03

2.4 0x42 Read Basic Information

The master sends a command:

Address code	1 byte	Slave address
Function code	1 byte	0x42
Number parameters	1 byte	0x09
Reserved	1 byte	0x00
Reserved	1 byte	0x00

E.g. command 01 42 09 00 00 BA DD is sent, the slave responds:

Address code	1 byte	Slave address
Function code	1 byte	0x42
Number of parameters	1 byte	0x09
The 1st parameter	2 bytes	Product model
The 2nd parameter	2 bytes	Cabinet sensor temperature
The 3rd parameter	2 bytes	Defrost sensor temperature
The 4th parameter	2 bytes	On temp in cooling mode
The 5th parameter	2 bytes	Off temp in cooling mode
The 6th parameter	2 bytes	On temp in heating mode
The 7th parameter	2 bytes	Off temp in heating mode
The 8th parameter	2 bytes	Alarm status
The 9th parameter	2 bytes	Device status

Detailed parameter information is as follows:

Description	Range	Default	Unit	Data Resolution	Yes/No sign
Product model	1 ~ 2 (1: EK3030E， 2: ECB-1000S)	1	℃	0.1/bit	No
Cabinet sensor temperature	-40℃ ~ 99℃		℃	0.1/bit	Yes
Defrost sensor temperature	-40℃ ~ 99℃		℃	0.1/bit	Yes

On temp in cooling mode	Off temp in cooling mode ~ 85℃	10.0 ℃	℃	0.1/bit	Yes
Off temp in cooling mode	-40℃ ~ On temp in cooling mode	-10.0 ℃	℃	0.1/bit	Yes
On temp in heating mode	-40℃ ~ Off temp in heating mode	-10.0 ℃	℃	0.1/bit	Yes
Off temp in heating mode	On temp in heating mode ~ 85℃	10.0 ℃	℃	0.1/bit	Yes

Description	Definition of bits			
Alarm status	MSByte	Bit7(MSb)	Reserved	
		Bit6	Reserved	
		Bit5	Reserved	
		Bit4	Reserved	
		Bit3	Reserved	
		Bit2	Reserved	
		Bit1	Reserved	
		Bit0(LSb)	Door opening alarm	0: No 1: Yes
	LSByte	Bit7(MSb)	Pressure switch alarm	0: No 1: Yes
		Bit6	Emergency external alarm	0: No 1: Yes
		Bit5	Normal external alarm	0: No 1: Yes
		Bit4	Over trial time alarm	0: No 1: Yes
		Bit3	Over lower limit alarm value	0: No 1: Yes
		Bit2	Over upper limit alarm	0: No 1: Yes
		Bit1	Defrost sensor fault alarm	0: No 1: Yes
		Bit0(LSb)	Cabinet sensor fault alarm	0: No 1: Yes
Device status	MSByte	Bit7(MSb)	Reserved	
		Bit6	Reserved	
		Bit5	Reserved	
		Bit4	Reserved	
		Bit3	Reserved	
		Bit2	Reserved	
		Bit1	Reserved	
		Bit0(LSb)	Reserved	
	LSByte	Bit7(MSb)	Reserved	
		Bit6	Reserved	
		Bit5	Reserved	
		Bit4	Cooling or heating status	0: Cooling 1: Heating
		Bit3	Fan status	0: Off 1: On
		Bit2	Heating on	0: Off 1: On
		Bit1	Cooling on	0: Off 1: On
		Bit0(LSb)	Defrost status	0: Off 1: On

Error

Device address	1 byte	Slave address
Function code	1 byte	0xC2(0x42 + 0x80)
Exception code	1 byte	01 or 02 or 03

3. REGISTER ADDRESS

3.1 Parameters (Read-only)

Read-only						
Register address	Description	Range	Default	Unit	Data Resolution	Yes/No sign
0x0100	Cabinet sensor temperature	-40℃ ~ 99℃		℃	0.1/bit	Yes
0x0101	Defrost sensor temperature	-40℃ ~ 99℃		℃	0.1/bit	Yes
0x0102	H18 (Software version)	0: V1.0	0		1/bit	

3.2 Parameters (Read/Write)

Read/Write							
Register address	Parameter code	Description	Range	Default	Unit	Data Resolution	Yes/No sign
0x0400		On temp in cooling mode	Off temp in cooling mode ~ 85℃	10.0 ℃	℃	0.1/bit	Yes
0x0401		Off temp in cooling mode	-40℃ ~ On temp in cooling mode	-10.0 ℃	℃	0.1/bit	Yes
0x0402		On temp in heating mode	-40℃ ~ Off temp in heating mode	-10.0 ℃	℃	0.1/bit	Yes
0x0403		Off temp in heating mode	On temp in heating mode ~ 85℃	10.0 ℃	℃	0.1/bit	Yes
0x0404	F1	Defrost time	1~120min	30min	min	1/bit	
0x0405	F2	Defrost cycle	0~120h	6h	h	1/bit	
0x0406	F3	Defrost cycle calculation	0: Cumulative controller work time after power on 1: Cumulative compressor work time after power on 2: Controlled by external defrost timer	1		1/bit	
0x0407	F4	Dripping time after defrost	0~120min	3 min	min	1/bit	

0x0408	F5	Defrost type	0: Electric heating defrost 1: Hot gas defrost 2: Wind defrost	0		1/bit	
0x0409	F6	Defrost stop temperature	-40.0℃~+50.0℃	10.0℃	℃	0.1/bit	Yes

0x040A	F7	Fan running mode	-180~-1: Fan starts ahead of compressor 180~1s 0: Fan and compressor run synchronically 1: Fan runs continuously (no start delay after dripping) 2: Fan runs continuously, stops during defrost and dripping 3~302: Fan starts 1~300s later than compressor	0		1/bit	Yes
0x040B	F8	Initial fan start delay after dripping	0~300s	30s	s	1/bit	
0x040C	F9	Compressor start delay	0~10min	0	min	1/bit	
0x040D	F10	Over temperature alarm delay after power on	0~24h	2h	h	1/bit	
0x040E	F11	Over temperature alarm	0~50.0℃	5.0℃	℃	0.1/bit	
0x040F	F12	Over temperature alarm delay	0~120min	10min	min	1/bit	
0x0410	F13	Cabinet sensor calibration temperature	-10.0℃~+10.0℃	0.0℃	℃	0.1/bit	Yes
0x0411	H1	Compressor stop time in the mode of “Run/stop in a proportional time”	1~60min	30min	min	1/bit	
0x0412	H2	Compressor	0~60min	15min	min	1/bit	

		start-up time in the mode of “Run/stop in a proportional time”					
0x0413	H3	Over upper limit alarm value	Over lower limit alarm value~85.0℃	20.0℃	℃	0.1/bit	Yes
0x0414	H4	Over lower limit alarm value	-40.0℃~Over upper limit alarm value	-20.0℃	℃	0.1/bit	Yes
0x0415	H5	Over-temperature alarm mode	0: Absolute temperature value 1: On/Off temp value ± Over-temperature alarm value	1		1/bit	
0x0416	H6	Whether to enable buzzer beep	0: No 1: Yes	1		1/bit	
0x0417	H7	Display mode in defrost and dripping (including the 15min after dripping)	0: Normal cabinet temperature 1: dEF 2: Cabinet temperature when defrosting starts	0		1/bit	
0x0418	H8	Whether to enable evaporator sensor	0: No 1: Yes	1		1/bit	
0x0419	H9	Temperature calibration of defrost sensor	-10.0℃~+10.0℃	0.0℃	℃	0.1/bit	Yes

0x041A	H10	Fan-managed mode	0: Fan is controlled by compressor time 1: Fan is controlled by defrost probe temperature 2: Fan is controlled by temperature difference between cabinet and defrost sensor.	0		1/bit	
0x041B	H11	Fan start-up temperature	Fan stop temperature	-5.0℃	℃	0.1/bit	Yes
0x041C	H12	Fan stop temperature	Fan start-up temperature~50.0℃	10.0℃	℃	0.1/bit	Yes
0x041D	H13	Fan start-up temperature difference	0℃~+50.0℃	5.0℃	℃	0.1/bit	
0x041E	H14	Integer/decimal switch	0: decimal 1: integer	0		1/bit	
0x041F	H15	Digital input 1 definition	±6: Start defrost ±5: Heating mode ±4: Normal external alarm ±3: Emergency external alarm ±2: Pressure switch alarm ±1: Door switch alarm 0: Disabled Positive number represents it defaults normally open, close valid; Negative number represents it defaults normally closed, open valid.	0		1/bit	Yes
0x0420	H16	Digital input 2 definition		0		1/bit	Yes
0x0421	H17	Controller ID address	0~128 (controller ID address; 0: Not connected to network)	0		1/bit	
0x0422	H19			0		1/bit	

		Enable copy card function	0: Disabled 1: Enabled				
0x0423	A1	Normal external alarm delay	0~240s	60s			
0x0424	A2	Whether to stop compressor, defrost and fan in normal alarm	0: No influence 1: Only compressor stops 2: Compressor defrost and fan stop.	0			
0x0425	A3	Emergency external alarm delay	0~240s	0			
0x0426	A4	Whether to stop compressor, defrost and fan emergency external alarm	0: No influence 1: Only compressor stops 2: Compressor defrost and fan stop.	0			
0x0427	A5	Door switch alarm delay	0~240s	60s			
0x0428	A6	Compressor and fan status when door open	0: No influence 1: Close fan 2: Close compressor 3: Close fan and compressor	3			

0x0429	A7	Fan stop time when door open	0: ON (Not stop) 1: OFF (Not start before door closed) 2~241: 1~240s (stop time)	40s			
0x042A	A8	Enable pressure switch auto reset	0: Disabled 1~5: Auto reset times	2			
0x042B	A9	Pressure switch auto reset delay	0~30min	3min			
0x042C		Set trial time	0~999 days	0	Day	1/bit	

3.3 Status (Read-Only)

Read-Only					
Register address	Description	Definition of bits			
0x0800	Relay output status	MSByte	Bit7(MSb)	Reserved	
			Bit6	Reserved	
			Bit5	Reserved	
			Bit4	Reserved	
			Bit3	Reserved	
			Bit2	Reserved	
			Bit1	Reserved	
			Bit0(LSb)	Reserved	
		LSByte	Bit7(MSb)	Reserved	
			Bit6	Reserved	
			Bit5	Reserved	
			Bit4	Reserved	
			Bit3	Reserved	
			Bit2	Fan relay	0: Close 1: Open
			Bit1	Heating wire relay	0: Close 1: Open
			Bit0(LSb)	Compressor relay	0: Close 1: Open

0x0801	Digital input switch status	MSByte	Bit7(MSb)	Reserved	
			Bit6	Reserved	
			Bit5	Reserved	
			Bit4	Reserved	
			Bit3	Reserved	
			Bit2	Reserved	
			Bit1	Reserved	
			Bit0(LSb)	Reserved	
		LSByte	Bit7(MSb)	Reserved	
			Bit6	Reserved	
			Bit5	Reserved	
			Bit4	Reserved	
			Bit3	Reserved	
			Bit2	Reserved	
			Bit1	DI2 digital input	0: No 1: Yes
			Bit0(LSb)	DI1 digital input	0: No 1: Yes
0x0802	Alarm status	MSByte	Bit7(MSb)	Reserved	
			Bit6	Reserved	
			Bit5	Reserved	
			Bit4	Reserved	
			Bit3	Reserved	
			Bit2	Reserved	
			Bit1	Reserved	
			Bit0(LSb)	Door opening alarm	0: No 1: Yes
		LSByte	Bit7(MSb)	Pressure switch alarm	0: No 1: Yes
			Bit6	Emergency external alarm	0: No 1: Yes
			Bit5	Normal external alarm	0: No 1: Yes
			Bit4	Over trial time alarm	0: No 1: Yes
			Bit3	Over lower limit alarm value	0: No 1: Yes
			Bit2	Over upper limit alarm	0: No 1: Yes
			Bit1	Defrost sensor fault alarm	0: No 1: Yes
			Bit0(LSb)	Cabinet sensor fault alarm	0: No 1: Yes

3.4 Status (Read/Write)

Read/Write					
Register address	Description	Definition of bits			
0x0A00	Device status	MSByte	Bit7(MSb)	Reserved	
			Bit6	Reserved	
			Bit5	Reserved	
			Bit4	Reserved	
			Bit3	Reserved	
			Bit2	Reserved	
			Bit1	Reserved	
			Bit0(LSb)	Reserved	
		LSByte	Bit7(MSb)	Reserved	
			Bit6	Reserved	
			Bit5	Reserved	
			Bit4	Reserved	
			Bit3	Reserved	
			Bit2	Fan status	0: Off 1: On
			Bit1	Cooling status	0: Off 1: On
			Bit0(LSb)	Defrost status	0: Off 1: On